Sensor fusion combines readings from multiple sensors to estimate a more accurate and reliable representation of the physical system.

One of the most widely used techniques for this is the Kalman filter, which is an algorithm that provides an optimal estimate of the system's true state over time by continuously updating predictions based on both model dynamics and new measurements. It effectively smooths out noise and gives more weight to reliable information as it becomes available.

In a digital twin of a smart building, for example, temperature sensors distributed across a room may all report slightly different readings due to calibration errors or local airflow differences. The Kalman filter can fuse these readings to produce a stable and accurate estimate of the true room temperature.

EXAMPLE: FUSING MULTIPLE TEMPERATURE SENSOR READINGS WITH A KALMAN FILTER

The Python code in listing 3.5 demonstrates how a simple Kalman filter can be used to fuse readings from five noisy IoT temperature sensors into a single, accurate estimate of the true room temperature.